LOLR Model and Sunspot Results

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Economic Environment

- State variables in period t: private debt b_t, LOLR debt l_t, growth state $g_t \in \{g_L, g_H\}$, and transitory shock ε_t.
- Output in repayment: $y_t = g_t \exp(\sigma_\varepsilon \varepsilon_t)$.
- Output in default: $y_t^D = \phi_{g_t} y_t$ with state-dependent output loss.
- Growth follows a Markov chain P_g; ε_t is iid with invariant probabilities π_ε.
- International lenders are risk-neutral with gross risk-free rate $1 + r^*$.

Government Problem in Repayment

Given current (b, l, g, ε) and no default, the sovereign chooses next debt (b', l'):

$$V^{ND}(b, l, g, \varepsilon) = \max_{b', l'} \left\{ u(c) \; + \; \beta g^{1-\gamma} \mathbb{E}\left[\max\{ V^{ND}(b', l', g', \varepsilon'), V^{D}(b', l', g', \varepsilon') \} \right] \right\}.$$

Consumption in repayment is

$$c = y - b - l + n(b', l', g) + n_l(l', g),$$

where

$$n(b', l', g) = \frac{g\, b'}{R(b', l', g)}, \qquad n_l(l', g) = \frac{g\, l'}{R_l^{ND}}.$$

Government Problem in Default

In default, private market issuance is shut down and the sovereign only chooses LOLR debt:

$$V^D(b,l,g,\varepsilon) = \max_{l'} \Big\{ u\Big(y^D - l + \frac{g\, l'}{R_l^D}\Big) \; + \; \beta g^{1-\gamma}\, \mathbb{E}\Big[\theta V^D(\tilde{b}, l', g', \varepsilon') \\ + (1-\theta) \max\{V^{ND}(\kappa\tilde{b}, l', g', \varepsilon'), V^D(\tilde{b}, l', g', \varepsilon')\}\Big] \Big\},$$

with transformed debt states

$$\tilde{b} = \frac{b}{g}, \qquad \kappa\tilde{b} \text{ after restructuring on re-entry.}$$

How Q and X Are Built

At current state $(b, \ell, g, \varepsilon)$:

$$d(b, \ell, g, \varepsilon) = \mathbf{1}\{V^{ND}(b, \ell, g, \varepsilon) < V^{D}(b, \ell, g, \varepsilon)\}.$$

Bond price is the payoff decomposition

$$Q(b, \ell, g, \varepsilon) = (1 - d) + d\, X(b, \ell, g, \varepsilon),$$

where:

- if no default $(d = 0)$, lenders receive 1;
- if default $(d = 1)$, lenders receive continuation value X.

Continuation value satisfies

$$X(b, \ell, g, \varepsilon) = \frac{1}{1 + r^*} \mathbb{E}\big[\theta X' + (1 - \theta)\big((1 - e')X' + e'\, \kappa\, Q'\big) \mid g\big],$$

with $e' = \mathbf{1}\{V^{ND}(\kappa b', \ell', g', \varepsilon') \geq V^{D}(b', \ell', g', \varepsilon')\}$.

So in the code, (Q, X) are obtained jointly as a fixed point given current policy rules and the default/re-entry decisions.

Pricing, Constraints, and LOLR Policy

- Debt price recursion implies $Q = (1-d) + dX$, with default indicator $d = \mathbf{1}\{V^{ND} < V^{D}\}$.
- Gross market schedule and implied private issuance:

$$R(b', l', g) = \frac{1 + r^*}{\mathbb{E}[Q']}, \qquad n(b', l', g) = \frac{g\, b'}{R(b', l', g)}.$$

- Endogenous default probability schedule:

$$p^{def}(b', l', g) = \mathbb{E}[d'].$$

- Feasible repayment choices must satisfy:

$$p^{def}(b', l', g) \leq \bar{p}_u, \quad n(b', l', g) \leq \bar{n}_g, \quad l' \leq \bar{l}^{ND}.$$

- In default, LOLR borrowing is capped by $l' \leq \bar{l}^{D}$.

Numerical Solution in the Code

The solver in `1period_lolr/src/solver_lolr.jl` iterates:

- Solve default value V^D and default-state LOLR policy l'_D.
- Update default decision d and re-entry decision $e = 1 - d$.
- Solve bond-price fixed point for continuation value X and price Q.
- Compute schedules (R, n, p^{def}).
- Update repayment value V^{ND} under feasibility constraints.
- Recover optimal policy indexes (b', l') for simulation.

Simulation in `simulation_lolr.jl` then generates moments, default episodes, and debt distributions.

Sunspot Outputs Used in This Section

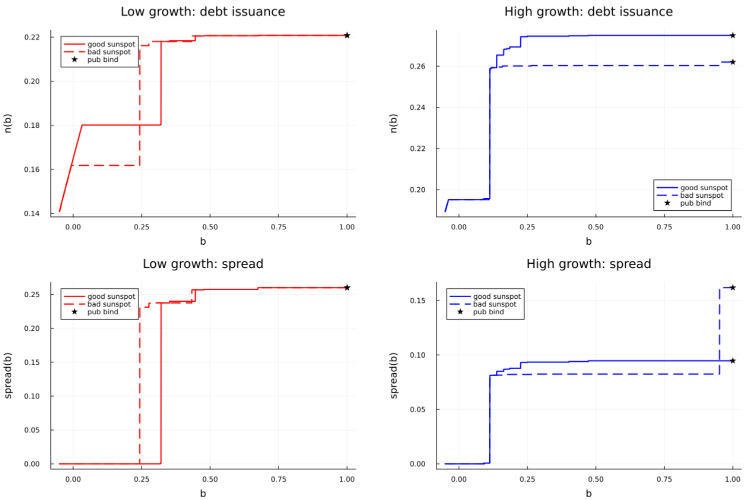
- Moments table file: `../1period_sunspot/result/sunspot_moments_pb.tex`.
- Policy figure file: `../1period_sunspot/result/sunspot_policy_2x2.png`.
- The table compares $P_b = 25\%$ vs. $P_b = 1\%$ for:
  - First moments.
  - Low-growth-state moments.
  - High-growth-state moments.

Sunspot Moments Table

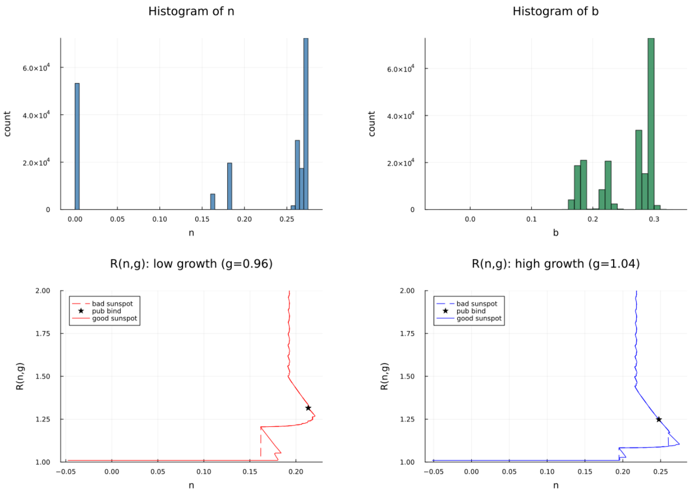
Table: Sunspot moments for alternative bad-sunspot probabilities (P_b)

Moment	P_b=25%	P_b=1%
First moments		
avg(spread)	0.0731	0.0706
avg(qb/y)	0.2455	0.2522
avg(f/y)	0.2670	0.2735
avg(n/y)	0.2455	0.2522
avg(b/y)	0.2411	0.2488
avg(tb/y)	-0.0044	-0.0034
default rate	0.1525	0.1442
Low-growth state		
avg(spread)	0.0000	0.0016
avg(qb/y)	0.1827	0.2110
avg(f/y)	0.1846	0.2135
avg(n/y)	0.1827	0.2110
avg(b/y)	0.1882	0.2100
avg(tb/y)	0.0055	-0.0010
default rate	0.3873	0.3691
High-growth state		
avg(spread)	0.0896	0.0874
avg(qb/y)	0.2597	0.2623
avg(f/y)	0.2856	0.2882
avg(n/y)	0.2597	0.2623
avg(b/y)	0.2531	0.2583
avg(tb/y)	-0.0067	-0.0040
default rate	0.0000	0.0000

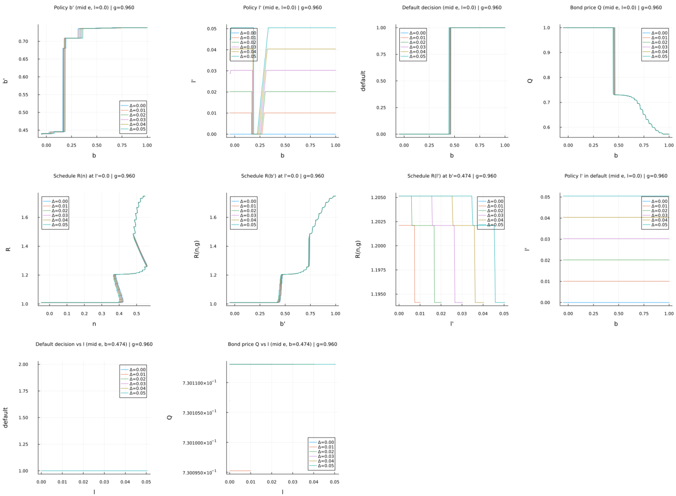
Sunspot Optimal Policy: 2x2 Figure



Sunspot n and R Figure



LOLR Policies by Δ (Spread 0.06)



LOLR Moments by Δ at Spread 0.06

Table: LOLR moments across Δ at fixed spread = 0.06

Moment	Δ=0.00	Δ=0.01	Δ=0.02	Δ=0.03	Δ=0.04	Δ=0.05
First moments						
avg(spread)	0.0921	0.0923	0.0921	0.0921	0.0921	0.0921
avg(qb/y)	0.6701	0.6701	0.6701	0.6701	0.6701	0.6701
avg(f/y)	0.7386	0.7386	0.7386	0.7386	0.7386	0.7386
avg(n/y)	0.6701	0.6701	0.6701	0.6701	0.6701	0.6701
avg(b/y)	0.6786	0.6786	0.6786	0.6786	0.6786	0.6786
avg(tb/y)	0.0085	0.0088	0.0091	0.0093	0.0096	0.0099
default rate	0.1480	0.1480	0.1480	0.1480	0.1480	0.1480
Low-growth state						
avg(spread)	0.2510	0.2510	0.2510	0.2510	0.2510	0.2510
avg(qb/y)	0.5833	0.5832	0.5833	0.5833	0.5833	0.5833
avg(f/y)	0.7355	0.7353	0.7355	0.7355	0.7355	0.7355
avg(n/y)	0.5833	0.5832	0.5833	0.5833	0.5833	0.5833
avg(b/y)	0.4220	0.4220	0.4220	0.4220	0.4220	0.4220
avg(tb/y)	-0.1614	-0.1601	-0.1592	-0.1581	-0.1571	-0.1560
default rate	0.3880	0.3880	0.3880	0.3880	0.3880	0.3880
High-growth state						
avg(spread)	0.0921	0.0922	0.0921	0.0921	0.0921	0.0921
avg(qb/y)	0.6702	0.6701	0.6702	0.6702	0.6702	0.6702
avg(f/y)	0.7386	0.7386	0.7386	0.7386	0.7386	0.7386
avg(n/y)	0.6702	0.6701	0.6702	0.6702	0.6702	0.6702
avg(b/y)	0.6788	0.6788	0.6788	0.6788	0.6788	0.6788
avg(tb/y)	0.0086	0.0089	0.0091	0.0094	0.0097	0.0099
default rate	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000